⚙️ Setup and Installation

Follow the steps below to build and run the **Smart Railway Platform Clock & Announcement Controller** project.

1. Install Required Software

Install the following software on your computer:

- **Keil µVision** – for Embedded C development and firmware compilation
- **Proteus Design Suite** – for circuit simulation
- **LPC2129 device support/package** – required for compiling the LPC2129 project

2. Clone or Download the Repository

Clone the GitHub repository using:

```
git clone <your-github-repository-url>
```

Or download the repository as a ZIP file and extract it.

Open the extracted project folder.

3. Open the Keil Project

Open **Keil µVision**.

Open the project file:

```
Smart-Railway-Platform-Clock.uvproj
```

or the corresponding `.uvprojx` file available in the project folder.

The project should contain all required source and header files.

4. Verify Project Files

Make sure the following important files are included:

```
main.c

lcd.c
lcd.h
```

```
lcd_defines.h

rtc.c
rtc.h

kpm.c
kpm.h

train_db.c
train_db.h

train_info_display.c
train_info_display.h

led.c
led.h

buzzer.c
buzzer.h

eint_interrupt.c
eint_interrupt.h

menu.c
menu.h

delay.c
delay.h

defines.h
types.h
```

5. Configure the LPC2129 Target

In Keil:

```
Project
   ↓
Options for Target
   ↓
Device
```

Select:

```
NXP LPC2129
```

Verify the target configuration before building the project.

6. Configure the Clock

Set the project clock configuration according to the crystal and clock settings used in the Proteus circuit.

Make sure the following are consistent:

- Crystal frequency
- CCLK configuration
- PCLK configuration
- RTC clock configuration

The firmware clock configuration and Proteus crystal frequency should match.

7. Configure GPIO Pins

Verify that the GPIO definitions in the source code match the Proteus circuit.

Important modules include:

```
LCD        → LPC2129 GPIO
Keypad     → LPC2129 GPIO
Green LED  → P0.23
Yellow LED→ P0.24
Red LED    → P0.25
Buzzer     → Configured GPIO
EINT       → External interrupt pin
```

If any hardware pin is changed, update the corresponding firmware configuration.

8. Configure the Train Database

Open:

```
train_db.c
```

Add or modify train information in the train database.

Example:

```
TB traindb[MAX] =
{
    {20701,"Tirupati Vande Bharat","GUNTUR",
     1,9,30,9,40,0,0},

    {17256,"Narasapur Express","LINGAMPALLI",
     2,10,30,10,40,0,0},

    {12747,"Palnadu Super Fast Express","VIKARABAD",
     3,11,0,11,10,0,0}
};
```

The database contains:

```
Train Number
Train Name
Destination
Platform Number
Arrival Hour
Arrival Minute
Departure Hour
Departure Minute
Delay Hour
Delay Minute
```

9. Configure RTC

The RTC must be initialized with the required time and date.

For testing, configure the RTC with a time close to one of the train schedules.

For example:

```
SetRTCTimeInfo(9,29,00);
```

If the train arrival time is:

```
09:30
```

the system can be tested by setting the RTC close to `09:30`.

10. Build the Project

In Keil, select:

```
Project
   ↓
Build Target
```

or press:

```
F7
```

Check the **Build Output** window.

The project should compile without errors.

11. Generate the HEX File

Enable HEX file generation in Keil:

```
Options for Target
   ↓
Output
   ↓
Create HEX File
```

Then rebuild the project.

The generated HEX file will normally be located inside the project's output/build directory.

Example:

```
Objects/
└── Smart-Railway-Platform-Clock.hex
```

12. Open the Proteus Project

Open the Proteus project:

```
Proteus/
└── Smart_Railway_Controller.pdsprj
```

Check that the following components are connected:

```
LPC2129
16×2 LCD
4×4 Keypad
LEDs
Buzzer
Crystal
Power Supply
External Interrupt Push Button
```

13. Load the HEX File into LPC2129

In Proteus:

1. Double-click the LPC2129.
2. Locate the **Program File** field.
3. Select the generated `.hex` file.
4. Set the processor clock frequency according to the project configuration.
5. Click **OK**.

14. Run the Simulation

Start the Proteus simulation.

The system should initialize and display the project information.

The LCD should then display the RTC information or train information depending on the current RTC time.

15. Test Train Display

Set the RTC close to a train arrival time.

For example:

```
Train Arrival Time = 09:30
Current RTC Time   = 09:30
```

The system should detect the train and display:

```
20701:Tirupati V
PL1 A09:30 D09:40
```

The train name will scroll according to the programmed display logic.

16. Test Train Status LEDs

Test the different train conditions.

Before Arrival

Approximately 5 minutes before arrival:

```
YELLOW LED = ON
```

On Platform

Between arrival and departure:

```
GREEN LED = ON
```

Delayed Train

When a delay is configured:

```
RED LED = ON
```

If:

```
Delay = 0
```

the red LED remains OFF.

17. Test Buzzer

The buzzer should activate according to the programmed alert conditions, such as:

- Train approaching
- Delay information updated

18. Test Administrator Mode

Press the external interrupt/admin button.

The system should enter administrator mode and request the password.

Example:

```
Enter Password
****
```

After successful authentication, the administrator can access the available editing functions.

19. Test Train Time Editing

From administrator mode:

```
Admin Menu
     ↓
Edit Train Time
     ↓
Enter Train Number
     ↓
Enter Arrival Time
     ↓
Enter Departure Time
     ↓
Update Database
```

The updated train timing is then used by the scheduler.

20. Test Train Delay Editing

From administrator mode:

```
Admin Menu
     ↓
Edit Train Delay
     ↓
Enter Train Number
     ↓
Enter Delay
```

```
      ↓
Update Delay
```

The updated delay should affect the train schedule and LED indication.

21. Verify RTC Display

When no train matches the current time, the LCD should return to the RTC display.

Example:

```
TIME:10:25:35
THU 20/08/2026
```

The RTC display should continuously update as the time changes.

🧪 Quick Testing Procedure

For a quick demonstration:

```
1. Open Keil project
        ↓
2. Build the project
        ↓
3. Generate HEX file
        ↓
4. Open Proteus
        ↓
5. Load HEX into LPC2129
        ↓
6. Set RTC time
        ↓
7. Run simulation
        ↓
8. Check LCD
        ↓
9. Check LEDs
        ↓
10. Check buzzer
        ↓
11. Test keypad
        ↓
12. Test Admin Mode
```

❗ Common Setup Problems

LCD is Blank

Check:

- LCD power supply
- LCD contrast
- RS/RW/EN connections
- D0–D7 connections
- GPIO configuration
- LCD initialization code

Train Information is Not Displayed

Check:

- RTC time
- Train database
- Arrival/departure time
- Delay value
- Train matching logic

LEDs Are Not Working

Check:

- GPIO direction configuration
- LED polarity
- Current-limiting resistor
- LED pin definitions
- Active-high/active-low logic

Buzzer is Not Working

Check:

- Buzzer GPIO configuration
- Buzzer polarity
- Driver circuit, if used
- Buzzer control logic

Keypad is Not Responding

Check:

- Row and column connections
- GPIO direction

- Keypad pin definitions
- Key scanning logic

Proteus Simulation is Not Running Correctly

Check:

- Correct LPC2129 device
- Correct HEX file
- Crystal frequency
- Power connections
- Common ground
- GPIO connections
- Keil build errors

✅ Final Verification

Before uploading the project to GitHub, verify:

- [ ] Keil project builds without errors
- [ ] HEX file is generated
- [ ] Proteus circuit opens correctly
- [ ] LCD displays correctly
- [ ] RTC works correctly
- [ ] Train matching works
- [ ] Train name scrolling works
- [ ] Keypad works
- [ ] Admin password works
- [ ] Train time editing works
- [ ] Train delay editing works
- [ ] Red LED works for delay
- [ ] Yellow LED works before arrival
- [ ] Green LED works while train is on platform
- [ ] Buzzer alert works